Individual-Level Evidence for Temporal Bayesian Player-Skill Estimation

CSE 516 – Probabilistic Graphical Models

M2 Extension Report | Group 3

Prepared by

Dhruvi Shah • Divija Nayak • Veera Bharakhada

This report documents our M2 extension to a temporal Bayesian player-skill estimation project. The selected limitation is that team-level win/loss outcomes alone may not provide enough evidence to distinguish the individual abilities of players on the same team. We investigate whether augmenting shared match outcomes with player-specific performance observations addresses this limitation.

Because the supplied ATP history dataset does not contain the individual match statistics required for this extension, the initial investigation uses clearly labelled synthetic doubles data. A simplified probabilistic model is fitted using maximum a posteriori (MAP) optimisation, with approximate uncertainty obtained through a Laplace approximation.

In the preliminary single-seed experiment, adding individual-level observations reduced skill MAE from 0.372678 to 0.218028 and skill RMSE from 0.466809 to 0.285278. Held-out log-loss decreased from 0.596247 to 0.554177, while Brier score decreased from 0.204032 to 0.186403.

These results demonstrate the proposed mechanism under the assumptions of the synthetic data generator; they do not establish real-tennis validation or multi-seed statistical evidence.

1. Project Context

Our project investigates Bayesian estimation of player skill as it changes over time. In a temporal probabilistic graphical model, each player's skill is represented as a latent variable evolving across successive time points. Match outcomes provide evidence about the relative strengths of the participating teams.

Our earlier M2 core prototype provided a technical foundation for temporal skill estimation. This extension focuses on a specific limitation in team-based settings: a shared team result may not provide enough player-specific information to distinguish the contributions of teammates.

2. Identified Research Limitation

When the only observation for a team match is the overall team result, the model receives a shared outcome for multiple players. A win indicates that one team performed better, but it does not uniquely identify how much each teammate contributed.

Players with different underlying abilities can therefore be difficult to distinguish using team outcomes alone. This issue is especially relevant to doubles, where two players jointly produce one match result. We investigate whether additional player-specific evidence can improve individual skill estimation.

3. Proposed Solution

We propose adding individual player-performance observations alongside the overall team result.

In a real tennis dataset, possible observations could include player-specific aces, winners, or other match statistics. These are examples of potential real-world measurements; they are not available in the supplied ATP history file used for this extension.

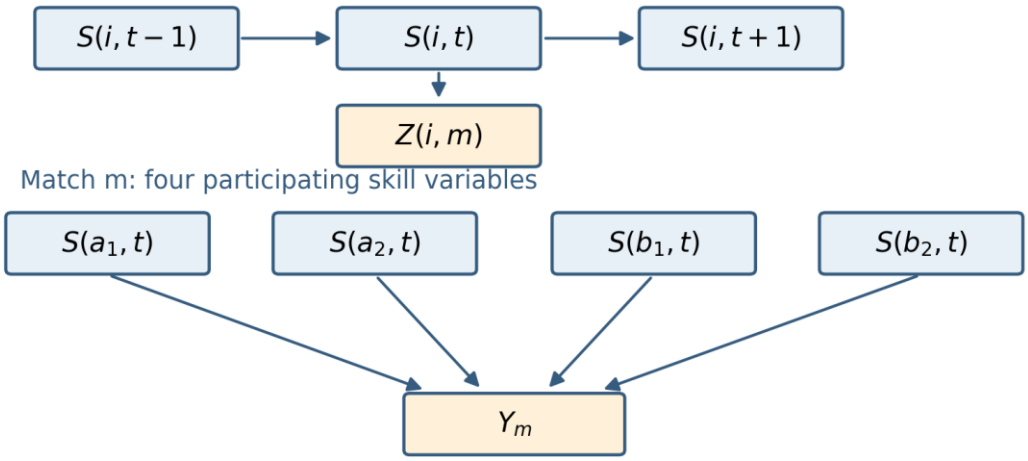
In our prototype, a synthetic individual observation is generated for each participating player as the player's latent skill plus Gaussian noise. This controlled setup tests whether additional player-specific evidence improves estimation when its relationship to latent skill is known.

Relationship to the M2 Core Implementation: We retain the temporal Bayesian skill-estimation framework as our conceptual foundation, but we implement this extension as a separate simplified probabilistic model. Our original M2 core prototype uses TTT-style expectation propagation, whereas our extension uses maximum a posteriori (MAP) optimisation and a Laplace approximation. We therefore investigate additional individual-level evidence rather than directly modifying the original expectation-propagation implementation.

4. Probabilistic Graphical Model Representation

Let S(i,t) denote the latent skill of player i at time t. Each player's skill forms a first-order temporal chain: skill at time t depends directly on skill at the preceding time point.

For match m at time t_m, let P_m denote the four participating players. The match outcome Y_m depends on the latent skills of all four participants. When individual evidence is available, observation Z(i,m) depends on the skill of player i for that match.



Note: Representative temporal chain for a player participating in match m.

Figure 1. Temporal Bayesian graphical model for individual-level evidence. Each player's latent skill evolves over consecutive time points. The match-outcome variable depends on the latent skills of all four participating players, while individual observations provide additional player-specific evidence. The figure illustrates a representative player's temporal dependencies; the complete model includes the skill chains of all players and the corresponding match factors.

5. Probabilistic Model

5.1 Initial Skill Prior

Each player's initial latent skill is assigned a Gaussian prior with mean zero and standard deviation 1.0:

$$S(i,0) \sim N(0, 1.0^2)$$

5.2 Temporal Skill Transition

Skill is assumed to evolve gradually through a Gaussian random walk. The temporal drift standard deviation is 0.12:

$$S(i,t) \mid S(i,t-1) \sim N(S(i,t-1), 0.12^2)$$

5.3 Team Match-Outcome Likelihood

For each doubles match, the probability that Team A wins is modelled using a logistic function of the difference between the teams' average latent skills.

Let A_m and B_m denote the two pairs and let T=0.8 denote the temperature parameter:

$$p(Y_m = 1 \mid S) = \sigma([mean(S_A_m) - mean(S_B_m)] / T)$$
$$T = 0.8$$

The match outcome is modelled as a Bernoulli variable with this probability. Its likelihood depends on the latent skills of all four participating players.

5.4 Individual-Observation Likelihood

When the extension is enabled, each available player-match observation is modelled as a noisy Gaussian measurement of that player's latent skill. The default observation standard deviation is 0.65:

$$Z(i,m) \mid S(i,t_{\mathrm{m}}) \sim N(S(i,t_{\mathrm{m}}), 0.65^2)$$

The team-outcome-only condition omits this observation likelihood. The individual observations are synthetic, not actual tennis statistics.

5.5 Joint Probability Structure

The joint probability distribution factorises into initial skill priors, temporal transition factors, match-outcome factors, and individual-observation factors:

$$\begin{aligned} p(S,Y,Z) = & \prod_i p\left(S_{i,0}\right) \\ & \times \prod_i \prod_{t=1}^{T-1} p\left(S_{i,t} \mid S_{i,t-1}\right) \\ & \times \prod_m p\left(Y_m \mid S_{\mathcal{P}_m,t_m}\right) \\ & \times \prod_{(i,m)\in\mathcal{O}} p\left(Z_{i,m} \mid S_{i,t_m}\right) \end{aligned}$$

Here, P_m denotes the four players participating in match m, t_m denotes the match time, and O denotes the set of available player-match observations.

The team-outcome-only condition omits the final product of individual-observation factors. Both conditions retain the same initial skill priors, temporal transitions, and match-outcome likelihoods.

The graphical structure is specified manually using chronological order and player participation; no structure-learning algorithm is applied.

6. Learning and Inference

Latent player-skill trajectories are inferred using maximum a posteriori (MAP) estimation. Given the available training-period evidence, the MAP estimate is:

$$\hat{S}_{\mathrm{MAP}} = \arg\max_S p(S \mid Y_{\mathrm{train}}, Z_{\mathrm{train}}) = \arg\min_S \mathcal{L}(S)$$

Here, L(S) is the negative log posterior, up to additive constants independent of S. For the team-outcome-only condition, Z_train is omitted.

The objective includes an initial Gaussian prior, Gaussian temporal transition penalties, logistic match-outcome losses, and Gaussian individual-observation penalties when the additional evidence is enabled.

We optimise the objective using SciPy's L-BFGS-B algorithm with an analytical gradient, a maximum of 500 iterations, and a relative function-tolerance setting of 10^-9.

The negative log-posterior objective, up to constants, is:

$$\begin{aligned} L(S) = & \sum_i S(i,0)^2 / (2{\sigma_0}^2) \\ & + \sum_i \sum_{t=1}^{T-1} [S(i,t) - S(i,t-1)]^2 / (2\sigma_d^2) \\ & + \sum_m BCE(Y_m, p_m(S)) \\ & + \sum_{(i,m)\in O} [Z(i,m) - S(i,t_m)]^2 / (2\sigma_z^2) \end{aligned}$$

The final summation is included only in the individual-evidence condition.

Approximate posterior uncertainty is obtained using a Laplace approximation around the MAP estimate. The inverse Hessian of the negative log posterior at the MAP solution approximates the posterior covariance; marginal standard deviations are obtained from its diagonal.

7. Dataset and Experimental Design

The supplied ATP history dataset provides context for our broader project and the separate reproduction of the original TTT baseline. It does not contain the individual match statistics required for this extension, so it is not used to evaluate the individual-evidence model.

We therefore generate synthetic doubles data with:

- 12 players
- 30 time points
- 12 matches per time point
- 360 matches in total
- Random seed 42

Each match contains two players per team. Player skills follow the Gaussian random walk described above. Team outcomes are sampled from the logistic probability based on the difference between average team skills. Each participating player receives a noisy synthetic individual-performance observation.

The dataset is split chronologically: the first 21 time points (70%) form the training period, and the final nine time points (30%) are reserved for evaluation.

During fitting, only training-period match outcomes and, in the extended condition, available training-period individual observations are used. Held-out match outcomes and individual observations are not incorporated into the MAP objective.

The model estimates latent skill trajectories over the full time horizon. Beyond the training period, the MAP estimates are determined by the Gaussian random-walk prior without updates from held-out match outcomes or individual observations. These estimates are compared against the known synthetic latent skills to calculate skill MAE and RMSE.

Held-out match outcomes are used only to evaluate predicted win probabilities through log-loss and Brier score.

This evaluation measures both recovery of simulated latent skills and prediction of unseen match outcomes under the specified synthetic data-generating process. The generated individual observations are not actual tennis performance statistics.

8. Evaluation Metrics

Four metrics are used to assess skill recovery and held-out match prediction in the preliminary experiment:

- **Skill MAE:** The mean absolute difference between estimated and known synthetic latent skills.
- **Skill RMSE:** The square root of the mean squared difference between estimated and known synthetic latent skills.
- **Log-loss:** The negative log-likelihood of held-out match outcomes under the predicted win probabilities; lower values indicate better probabilistic predictions.
- **Brier score:** The mean squared error between predicted probabilities and binary held-out outcomes; lower values indicate better probability estimates.

9. Initial Single-Seed Results

Table 1 reports results from our initial synthetic demonstration using random seed 42. The same generated matches and chronological split are used in both conditions.

Metric	Team outcome only	Team + individual evidence
Skill MAE	0.372678	0.218028
Skill RMSE	0.466809	0.285278
Held-out log-loss	0.596247	0.554177
Held-out Brier score	0.204032	0.186403

Table 1. Preliminary single-seed results for the team-outcome-only condition and the condition augmented with synthetic individual-level evidence.

10. Interpretation of Results

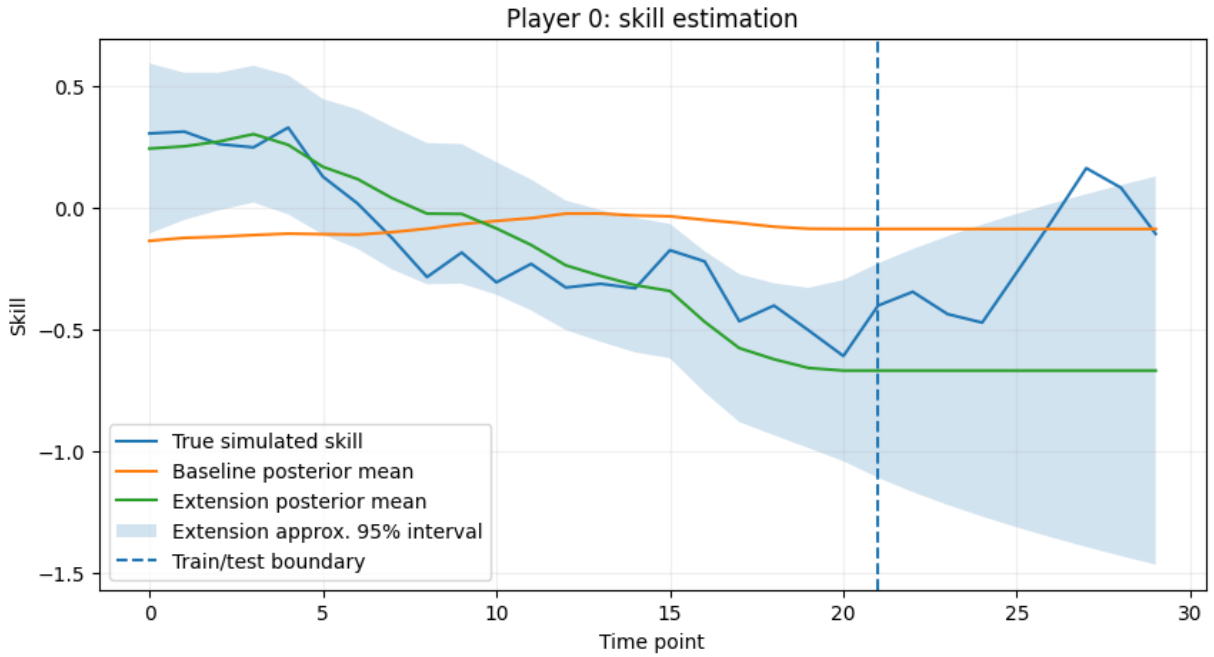


Figure 2. Player 0 skill-estimation trajectories in the synthetic doubles experiment (seed 42). The figure compares the true simulated skill with posterior mean estimates from the team-outcome-only baseline and individual-evidence extension. The shaded region represents the extension's approximate 95% posterior interval, and the dashed vertical line marks the chronological training/testing boundary.

The extension follows the true simulated skill more closely during much of the training period, whereas the baseline estimate remains relatively flat. Beyond the training boundary, the estimated skill stays approximately constant and posterior uncertainty increases because held-out observations are not used for updating. This plot illustrates one player and does not imply improvement for every player at every time point.

In this controlled synthetic demonstration, adding individual observations reduced both skill-estimation errors and held-out team-outcome probability scores.

Skill MAE decreased from 0.372678 to 0.218028, while RMSE decreased from 0.466809 to 0.285278. Log-loss and Brier score also decreased.

This pattern is consistent with our proposed mechanism: player-specific observations provide information that is not contained in a shared team result.

The findings should be interpreted within the assumptions of the synthetic data generator. Individual observations were generated directly from latent skill plus Gaussian noise, making them intentionally informative about the quantity being estimated. The experiment demonstrates that the model can use such evidence in this controlled setting; it does not establish that real tennis statistics have the same direct relationship with latent skill.

11. Limitations and Threats to Validity

The dataset and individual observations are synthetic. The observation model assumes a direct Gaussian relationship between each individual observation and latent skill.

Real statistics such as aces, winners, or unforced errors may also depend on opponent strength, surface, playing style, match duration, tactics, and other contextual factors. A more realistic model would need to account for these effects.

Our extension is a simplified temporal Bayesian model using MAP estimation and a Laplace approximation; it is not a reproduction of the complete published TrueSkill Through Time expectation-propagation implementation.

The initial demonstration uses one random seed, so the reported values do not establish statistical consistency. Held-out evaluation is based on a synthetic data-generating process and cannot substitute for validation using real player-specific performance measurements.

12. Conclusion and Future Direction

In this M2 extension, we identified a limitation of relying only on team-level match outcomes for estimating individual player skill in doubles. We proposed and implemented an additional individual-level observation model and demonstrated its use in a simplified temporal Bayesian framework.

In the preliminary single-seed synthetic experiment, the additional evidence was associated with lower skill-estimation errors and improved held-out probability scores.

In the next stage of our project, we plan to extend this preliminary single-seed demonstration into a systematic evaluation across multiple random seeds. The planned analysis will examine statistical consistency, sensitivity to individual-observation noise, robustness to missing observations, and the effect of match-data availability. These experiments will help determine whether the observed improvement is consistent under the assumptions of the synthetic model. Validation using real tennis data remains a longer-term direction requiring suitable player-specific performance measurements.

References

[1] Landfried, K. and Mocskos, E. (2025). “TrueSkill Through Time: Reliable Initial Skill Estimates and Historical Comparability with Julia, Python, and R.” *Journal of Statistical Software*, 112(6). https://doi.org/10.18637/jss.v112.i06.

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[3] Group 3. *CSE516_Limitation_and_Solution.ipynb*. M2 extension implementation and preliminary single-seed experiment, CSE 516 Probabilistic Graphical Models, 2026.